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Although there is a link between the different measures of information presented, these measures in fact measure different concepts of information. Both the Kullback-Leibler and the Shannon-Lindley measures are measures of the extent to which the observations support the prior distribution as a whole. The average information, which is the same in either case, then measures the average support. It is this average support that is symmetric with respect to the distribution of the parameters (the prior) and the distribution of the observations. The average amount of information in the experiment may therefore be considered as the mutual information of the parameters and the observations. As pointed out by Lindley (1956), it provides knowledge about the state of nature. The authors provide an interesting generalization to the case of more than one parameter.

Fisher's information on the other hand, provides a measure of information supplied by the data about an unknown parameter. In the Bayesian context, it may be informative to switch the Fisher information operator around, by observing the amount of information that the value of the parameter supply about the data. If X is an estimator of θ , then the Fisher information of θ with respect to the observed value x of X can be defined as :

$$FI = E_{\theta/x} \left\{ \frac{\partial \ln \Pi(\theta/x)}{\partial x} \right\}^2$$

Consider the normal model discussed in the paper (see Lindley (1956 p. 1000)) where x is normally distributed with mean θ and known variance σ^2 . The prior distribution of θ is normal with mean μ and variance τ^2 . The average amount of information in the experiment is

$$I(\epsilon, \Pi(\theta)) = I_x = \frac{1}{2} \ln(1 + \tau^2/\sigma^2).$$

The Fisher information in θ with respect to x is given by:

$$FI = -E \frac{\partial^2 \ln f(\theta/x)}{\partial x^2}$$
$$= \frac{\tau^2/\sigma^2}{1 + \tau^2/\sigma^2} \cdot \frac{1}{\sigma^2}$$

The information in the experiment is just a function of τ^2/σ^2 , while the Fisher information is also a decreasing function of σ^2 . Moreover, $I(\epsilon, \Pi(\theta))$ is an unbounded increasing function of τ , while FI is a bounded function of τ .

REFERENCE

LINDLEY, D.V. (1956). On a measure of the information provided by an experiment. *Ann. Math. Statist.*, **27**, 986-1005.